

Force

In physics, a **force** is a push or pull that acts on an object as a result of its interaction with another object.

It is an external agent capable of changing an object's state of rest or motion—causing it to accelerate, decelerate, or change direction.

Key Characteristics

- **Vector quantity** – forces have both magnitude (strength) and direction.
- **Standard unit** – the Newton (N).
- $1 \text{ N} = 1 \text{ kg} \cdot \text{m s}^{-2}$
- Equivalent to the force required to accelerate a 1 kg mass at a rate of 1 m s^{-2} .
- **Newton's Second Law** – the most common way to calculate force:

\[

$$\boxed{F = m \cdot a}$$

\]

where

* F = force (N)

* m = mass (kg)

* a = acceleration (m s^{-2})

Common Types of Forces

Contact Forces

These are applied through direct physical contact between objects. Examples include:

- **Friction** – resistance that opposes relative motion.
- **Tension** – pulling force transmitted through a string, rope, or cable.
- **Air resistance** – drag force exerted by the air on a moving object.
- **Push/Pull** – any direct mechanical action applied to an object.

(Non-contact forces such as gravity, electromagnetism, and nuclear forces are also fundamental, but are not listed in the OCR excerpt.)

Quick Reference

| Concept | Detail |

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| **Definition** | Push or pull due to interaction |

| **Vector** | Magnitude + direction |

| **Unit** | Newton (N) |

| **Formula** | $(F = m a)$ |

| **Contact examples** | Friction, tension, air resistance, push/pull |

Feel free to use these notes as a concise study guide for understanding the basics of force in physics.